SMART INDIA HACKATHON 2026 | PS 26078 (MOES / NCMRWF) | TEAM ECHO CRYPT

Megh-Drishti: Slide-by-Slide Storytelling Script

Professional spoken script designed to hook the jury with real disaster data, strong analogies, and zero boring jargon.

SLIDE 1 Title Slide & The Problem Hook

Target Time: 60 Seconds

THE STORY ANGLE: A SHOCKING QUESTION & DISASTER REALITY

Start with a question that makes judges look up from their scoring sheets immediately. Anchor the problem to a real historical disaster (Cyclone Amphan).

SPOKEN SCRIPT (WORD-FOR-WORD)

"Good morning, respected judges. Let me begin with a crucial question:

In May 2020, during Cyclone Amphan, thousands of families across coastal Bengal and Odisha had barely a few hours to evacuate before localized torrential rain submerged their villages. Conventional weather forecasts saw the storm coming, but they predicted broad regional rain—not the localized, catastrophic 150 mm/hr downpour that actually hit.

What if our disaster rescue forces had 5 to 10 days of clear warning—not for an entire state or district, but narrowed down to a precise 5-kilometer radius?

We are **Team Echo Crypt**, and today we present **Megh-Drishti (मेघ-दृष्टि)**: an AI-driven system for 3 to 10-day extreme weather tracking and precision downscaling, built for the Ministry of Earth Sciences and NCMRWF under Problem Statement 26078."

💡 **Speaker Delivery Tip:**

Pause for 2 seconds after asking the bold question. Make eye contact with the judges to let the magnitude of the problem sink in before introducing the team name.

SLIDE 2 Proposed Solution (Megh-Drishti: The Big Picture)

Target Time: 75 Seconds

THE STORY ANGLE: THE "CAMERA LENS" ANALOGY

Explain the 12 km grid defect simply: a blurred camera photo that needs a smart high-definition zoom lens.

SPOKEN SCRIPT (WORD-FOR-WORD)

"To understand why forecasts fail us today, imagine looking at a photo that has been blurred. Current global weather models run on a 12 km grid. In simple terms, a single 12 km box covers 144 square kilometers. If an intense storm hits a small village, the model averages it across the entire 144 km area. The peak disappears, the warning stays green or yellow, and disaster strikes unprepared.

Megh-Drishti solves this with a **Two-Stage AI approach** that acts like a smart zoom lens:

- **First:** Our **4D Spatio-Temporal Tracker** watches the atmosphere on a 3D spherical mesh. It calculates an **Extreme Forecast Index (EFI)** to spot anomalies 3 to 10 days ahead, tracking where the storm is headed with an expanding confidence cone.
- **Second:** Once an anomaly is detected, our **Generative Diffusion Model (DDPM)** zooms into that specific danger zone, sharpening the forecast from **12 km down to 5 km**—recovering the extreme rainfall peaks that normal models smooth out.
- **Finally:** This feeds into an interactive **GIS Command Centre** with live radar, and auto-generates **IMD-compliant 4-tier color alerts** that work even in network-dead zones via an offline-first PWA."

Transition to Next Slide: "Now, let's take a look under the hood to see how this technical engine actually works."

SLIDE 3 Technical Approach (Under the Hood)

Target Time: 75 Seconds

THE STORY ANGLE: AN ASSEMBLY LINE OF 4 CLEAN STEPS

Don't drown them in raw math equations; explain the flow as an assembly line that connects satellite data directly to rescue workers.

SPOKEN SCRIPT (WORD-FOR-WORD)

"Our technical pipeline is divided into four clean stages:

• **Stage 1 is Data Ingestion:** We take raw 12 km ensemble forecasts from NCMRWF and compare them against 30 years of climate history from ERA5. If an anomaly exceeds a standardized Z-score of +2.0, the pipeline triggers.
• **Stage 2 is 4D Atmospheric Tracking:** Instead of flat 2D maps that distort the Earth's poles, we use a **Spherical Graph Neural Network** on an icosahedral mesh. It tracks wind shear and pressure levels from the ground up to jet stream levels, outputting p10, p50, and p90 uncertainty bounds over 10 days.
• **Stage 3 is Generative Downscaling:** To sharpen 12 km into 5 km, simple interpolation blurs peaks, and standard AI hallucinates rain. We use a **Conditional Diffusion Model** paired with a 30-meter elevation map. Crucially, we enforce a **mass conservation rule**: the AI cannot create water that doesn't exist in the atmospheric column.
• **Stage 4 is Automated Alert Dispatch:** The output is evaluated against IMD thresholds and packaged into international **Common Alerting Protocol (CAP v1.2)** format.

Our frontend is built on **Next.js 16 and Leaflet** for 60-FPS map performance, backed by a lightning-fast **FastAPI** microservice."

Transition to Next Slide: "A great model on paper means nothing if it cannot survive the harsh reality of a disaster zone. Here is how we ensure feasibility."

SLIDE 4 Feasibility and Viability (Real-World Resilience)

Target Time: 70 Seconds

THE STORY ANGLE: DISARMING THE JUDGES' DOUBTS BEFORE THEY ASK

Target the 3 big questions judges always have: Big data speed, network collapse in floods, and mountain false alarms.

SPOKEN SCRIPT (WORD-FOR-WORD)

"Judges often ask: '*Can this actually run in government workflows, and what happens when disasters strike?*' We designed Megh-Drishti around three real-world bottlenecks:

• **Challenge 1: Weather files are massive (Gigabytes of data).**
Our Solution: We use cloud-native **Zarr and Dask** chunking. Instead of downloading full multi-gigabyte files, our backend streams only the spatial bounding box needed, delivering inference in just **340 milliseconds** on standard GPU servers.

• **Challenge 2: Severe storms knock down mobile towers.**
Our Solution: If an NDRF field officer loses mobile connectivity, the app does not stop. Built as an **Offline-First PWA**, the last 72 hours of forecast data remain accessible locally in **IndexedDB**. If an officer logs a damaged embankment, it queues in the device outbox and auto-syncs the moment 2G or satellite signal flickers back on.

• **Challenge 3: Mountain terrains trigger false alarms.**
Our Solution: By checking every forecast against **30 years of localized climate records**, normal monsoon rain over hilly terrain is not falsely flagged as an anomaly."

Transition to Next Slide: "So, what does all this technology mean for the people on the ground?"

SLIDE 5 Impact and Benefits (The Human Story)

Target Time: 75 Seconds

THE STORY ANGLE: FROM PANIC TO PREPARATION & LIVELIHOODS

Shift focus from code and equations to real lives saved, farmer crop security, and government cost savings.

SPOKEN SCRIPT (WORD-FOR-WORD)

"The true success metric of Megh-Drishti is not just accuracy percentages—it is the human and economic impact:

• **From Panic to Preparation:** Instead of a 3-hour panic response, authorities get **72 to 240 hours of strategic lead time**. NDRF teams can pre-position rescue boats in the exact 5 km zones most at risk.
• **Fewer False Evacuations:** When you issue a blanket warning for an entire district, people suffer 'warning fatigue' and ignore alerts. By narrowing warnings to a **5 km radius**, safe neighborhoods remain undisturbed while dangerous zones evacuate safely.
• **Protecting Agricultural Livelihood:** When farmers know 5 days in advance that their specific taluk faces severe convective rain, they can harvest early, move livestock to higher ground, and clear drainage channels.
• **Cost Efficiency:** Running full supercomputer numerical simulations at 5 km resolution costs crores in computing power. Our AI downscaler delivers equivalent precision in sub-second inference time using open-source, indigenous models."

Transition to Next Slide: "Everything we built is rooted in peer-reviewed science and official government standards."

SLIDE 6 Research and References (Scientific Grounding & Close)

Target Time: 60 Seconds

THE STORY ANGLE: ESTABLISHING 100% SCIENTIFIC AUTHORITY

Show that you didn't invent arbitrary formulas—you are using the exact same literature trusted by ECMWF, WMO, and MoES.

SPOKEN SCRIPT (WORD-FOR-WORD)

"Megh-Drishti is built on the shoulders of leading atmospheric and deep-learning research:

• **Atmospheric Dynamics:** We draw inspiration from Graph Neural Network climate modeling by *Lam et al. (Science 2023)* and *Keisler*.
• **Generative Downscaling:** Our diffusion formulation builds upon physical precipitation modeling by *Leinonen et al.* and *Mardani et al.*.
• **Extreme Forecast Index:** We strictly adopt the mathematical formulations established by *Lalaurette* and *ECMWF*, keeping EFI strictly on a valid **[-1.0, +1.0]** scale.
• **Official Government Data:** Trained and calibrated using **NCMRWF NEPS-G**, **Copernicus CDS ERA5**, and **IMD observational grids**.
• **Global Standards:** Full compliance with **WMO Multi-Hazard Early Warning Systems (MHEWS)** and the **Common Alerting Protocol (CAP)**.

With Megh-Drishti, we are turning coarse weather data into actionable, hyperlocal life-saving intelligence. Thank you, and we welcome your questions!"

💡 **Final Close Tip:**
End with a firm, energetic smile: "*Thank you, and we look forward to your cross-examination!*" Then switch the screen to the live dashboard on localhost:3000.